

REEF LIFE

**REEF FISH**

A huge variety of colorful fish live on the reefs. Some feed on coral or plantlike algae, while others prey on the many animals that live alongside them. Coral reefs support more fish species than any other ocean habitat, each specialized for its own way of life. Yet the total mass of fish is never as great as in the dense fish shoals of more nutrient-rich waters.

**SHRIMP**

The reefs swarm with shellfish and other invertebrates, including marine snails, colorful sea slugs, sea urchins, and crabs. Cleaner shrimp make a living by picking parasites off the skin, and even gills, of reef fish that queue up for treatment like cars at a busy car wash. Although many of these fish may eat other types of small animals, they never eat the cleaner shrimp.

**GIANT CLAM**

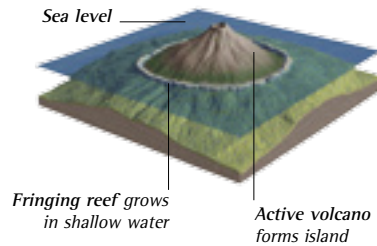
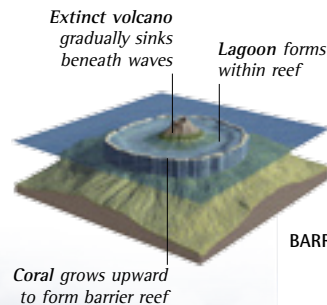
These enormous mollusks live in crevices in the reef, where they may grow to 5 ft (1.5 m) across. They catch plankton by filter-feeding, but like the reef corals they get most of their energy from sugar made by the zooxanthellae living in their colorful soft tissues. The biggest giant clams are unable to close their shells fully, and cannot grab divers' legs as people once believed.

**REEF SHARK**

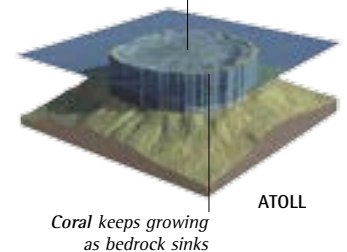
The reefs are patrolled by big predators like this reef shark, which hunt for small fish and other animals. They usually stay in the deeper water outside the reef face, seizing anything that ventures beyond the safety of the coral, but sometimes they swim into the shallow lagoon through gaps in the reef. Many hunt at night, using highly tuned senses to locate their prey in the dark.

► **CORAL ATOLLS**

The tropical Pacific is dotted with volcanic islands that have fringing reefs. When the volcanoes stop erupting, the islands gradually sink under their own weight, but their reefs keep growing upward. They become barrier reefs around shrinking central islands, and finally develop into ring-shaped atolls around roughly circular lagoons.

**FRINGING REEF****BARRIER REEF**

Volcanic peak vanishes beneath circular lagoon

**ATOLL****BARRIER REEF**◀ **LAGOONS AND ISLANDS**

The shallow, sun-warmed lagoon waters can reach temperatures of 95°F (35°C) or more. This is too hot for corals, but mobile animals, like these sharks, feed in the water when it is cooler. Low, palm-crowned islands that were built up from coral debris and sand are used as nesting sites by green turtles and frigate birds.

CROWN-OF-THORNS STARFISH

The crown-of-thorns starfish eats corals, turning its stomach inside out to engulf its prey. It can multiply into hungry swarms that wipe out large areas of coral, leaving dead white coral rock. But these starfish plagues are usually followed by population crashes when the starfish die off, and this gives the coral a chance to recover.

